AERIS Expert Review Packet

Anomaly

Source / metric	openaq / ozone
Observed / expected	0.081 / 0.015451
Time	2026-07-22 19:00 UTC
Location	29.6256, -95.2672
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	8 / 823	29.01 to 100; mean 71.514	37.58 at 2026-07-22 18:55Z; dt 5 min
asos	temperature	degC	8 / 823	23.2222 to 39; mean 29.9457	38 at 2026-07-22 18:55Z; dt 5 min
asos	wind_direction	degree	8 / 1030	0 to 360; mean 181.757	310 at 2026-07-22 18:55Z; dt 5 min
asos	wind_speed	m/s	8 / 1054	0 to 8.74555; mean 2.991	3.08666 at 2026-07-22 18:55Z; dt 5 min
noaa_gfs	gh_500	m	11 / 132	5915.17 to 5959.62; mean 5934.49	5938.94 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	pbl_height	m	11 / 132	17.7446 to 2110.42; mean 657.262	2017.14 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	precipitable_water	mm	11 / 132	29.7729 to 67.4723; mean 49.2857	55.1554 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	surface_pressure	hPa	11 / 132	1002.85 to 1014.29; mean 1008.24	1008.33 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	t_850	degC	11 / 132	17.875 to 24.7709; mean 21.3002	21.9881 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	u_10m	m/s	11 / 132	-4.5306 to 4.0994; mean 0.7294	0.231985 at 2026-07-22 18:00Z; dt 60 min
noaa_gfs	v_10m	m/s	11 / 132	-6.0398 to 5.78799; mean -0.4468	-2.53513 at 2026-07-22 18:00Z; dt 60 min
openaq	ozone	ppm	17 / 1139	-0.004 to 0.092; mean 0.0276	0.081 at 2026-07-22 19:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 178	3 to 98; mean 37.0449	73 at 2026-07-22 19:00Z; dt 0 min
openaq	pm25	ug/m3	77 / 2939	0 to 34.1; mean 9.3901	7.9 at 2026-07-22 19:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 360	0 to 100; mean 44.5639	72 at 2026-07-22 19:01Z; dt 1.9 min
openweather	humidity	percent	5 / 360	40 to 96; mean 72.7056	50 at 2026-07-22 19:01Z; dt 1.9 min
openweather	precipitation	mm/h	5 / 360	0 to 31.24; mean 0.3544	0 at 2026-07-22 19:01Z; dt 1.9 min
openweather	pressure	hPa	5 / 360	1007 to 1015; mean 1010.51	1009 at 2026-07-22 19:01Z; dt 1.9 min
openweather	temperature	degC	5 / 360	24.14 to 40.07; mean 30.7252	38.79 at 2026-07-22 19:01Z; dt 1.9 min
openweather	wind_direction	degree	5 / 360	0 to 360; mean 183.733	254 at 2026-07-22 19:01Z; dt 1.9 min
openweather	wind_speed	m/s	5 / 360	0.4 to 7.61; mean 2.4201	0.89 at 2026-07-22 19:01Z; dt 1.9 min
purpleair	pm25	ug/m3	69 / 4962	0 to 149.3; mean 11.5612	6.8 at 2026-07-22 19:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; dt 14.2 min
sentinel5p	s5p_ch4_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; dt 14.2 min
sentinel5p	s5p_co_column	mol/m^2	6 / 6	0.0233633 to 0.0312825; mean 0.0271	0.0312825 at 2026-07-22 18:45Z; dt 14.2 min
sentinel5p	s5p_co_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; dt 14.2 min
sentinel5p	s5p_hcho_column	mol/m^2	5 / 5	0.000163343 to 0.000396712; mean 0.0002	0.000396712 at 2026-07-22 19:14Z; dt 14.6 min
sentinel5p	s5p_hcho_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-22 19:14Z; dt 14.6 min
sentinel5p	s5p_no2_column	mol/m^2	2 / 2	5.0358e-05 to 5.87309e-05; mean 0.0001	5.0358e-05 at 2026-07-22 19:14Z; dt 14.6 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-07-22 19:14Z; dt 14.6 min
sentinel5p	s5p_o3_granule_available	count	8 / 8	1 to 1; mean 1	1 at 2026-07-22 18:45Z; dt 14.2 min
sentinel5p	s5p_so2_column	mol/m^2	5 / 5	-2.41161e-06 to 9.61895e-05; mean 0	-2.41161e-06 at 2026-07-22 19:14Z; dt 14.6 min
sentinel5p	s5p_so2_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-22 19:14Z; dt 14.6 min
tceq	co	ppm	3 / 195	0 to 0.5; mean 0.1436	0.1 at 2026-07-22 19:00Z; dt 0 min
tceq	no2	ppb	12 / 714	-2.6 to 44.3; mean 6.3	5.9 at 2026-07-22 19:00Z; dt 0 min
tceq	so2	ppb	3 / 191	-1 to 1.6; mean -0.2068	-0.4 at 2026-07-22 19:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	07-21 06Z 93.85, 12Z 74.67, 18Z 40.07, 07-22 00Z 62.73, 06Z 79.9, 12Z 60.19, 18Z 42.8, 07-23 00Z 69.27, 06Z 75.52, 12Z 74.08, 18Z 85.94, 07-24 00Z 93.19, 06Z 88.5
asos	temperature	07-21 06Z 24.96, 12Z 28.9, 18Z 35.38, 07-22 00Z 30.39, 06Z 26.94, 12Z 31.66, 18Z 37.75, 07-23 00Z 31.57, 06Z 29.18, 12Z 29.32, 18Z 28.22, 07-24 00Z 25.96, 06Z 25.77
asos	wind_direction	07-21 06Z 192.7, 12Z 259.7, 18Z 235.2, 07-22 00Z 194.3, 06Z 242.2, 12Z 287.5, 18Z 216.8, 07-23 00Z 150, 06Z 55.8, 12Z 52, 18Z 161.4, 07-24 00Z 152.2, 06Z 116.2
asos	wind_speed	07-21 06Z 2.044, 12Z 3.515, 18Z 2.448, 07-22 00Z 2.274, 06Z 2.882, 12Z 3.687, 18Z 2.709, 07-23 00Z 2.189, 06Z 3.052, 12Z 4.624, 18Z 3.022, 07-24 00Z 3.377, 06Z 2.414
noaa_gfs	gh_500	07-21 12Z 5917, 18Z 5932, 07-22 00Z 5930, 06Z 5935, 12Z 5927, 18Z 5940, 07-23 00Z 5930, 06Z 5943, 12Z 5927, 18Z 5934, 07-24 00Z 5940, 06Z 5957
noaa_gfs	pbl_height	07-21 12Z 199.6, 18Z 1199, 07-22 00Z 809.7, 06Z 112.4, 12Z 218.8, 18Z 1759, 07-23 00Z 530, 06Z 359.1, 12Z 342.8, 18Z 1310, 07-24 00Z 517.1, 06Z 530
noaa_gfs	precipitable_water	07-21 12Z 31.81, 18Z 35.48, 07-22 00Z 39.95, 06Z 45.28, 12Z 52.11, 18Z 55.5, 07-23 00Z 59.83, 06Z 46.46, 12Z 49.17, 18Z 60.98, 07-24 00Z 64.55, 06Z 50.31
noaa_gfs	surface_pressure	07-21 12Z 1010, 18Z 1010, 07-22 00Z 1007, 06Z 1008, 12Z 1008, 18Z 1008, 07-23 00Z 1005, 06Z 1007, 12Z 1007, 18Z 1007, 07-24 00Z 1008, 06Z 1012
noaa_gfs	t_850	07-21 12Z 22.43, 18Z 22.36, 07-22 00Z 21.93, 06Z 22.53, 12Z 21.73, 18Z 21.7, 07-23 00Z 24.01, 06Z 21.66, 12Z 20.17, 18Z 19.08, 07-24 00Z 19.89, 06Z 18.11
noaa_gfs	u_10m	07-21 12Z 2.627, 18Z 2.435, 07-22 00Z 1.136, 06Z 2.366, 12Z 3.039, 18Z 0.6747, 07-23 00Z 0.4549, 06Z -1.038, 12Z -0.5027, 18Z -1.79, 07-24 00Z 0.4981, 06Z -1.147

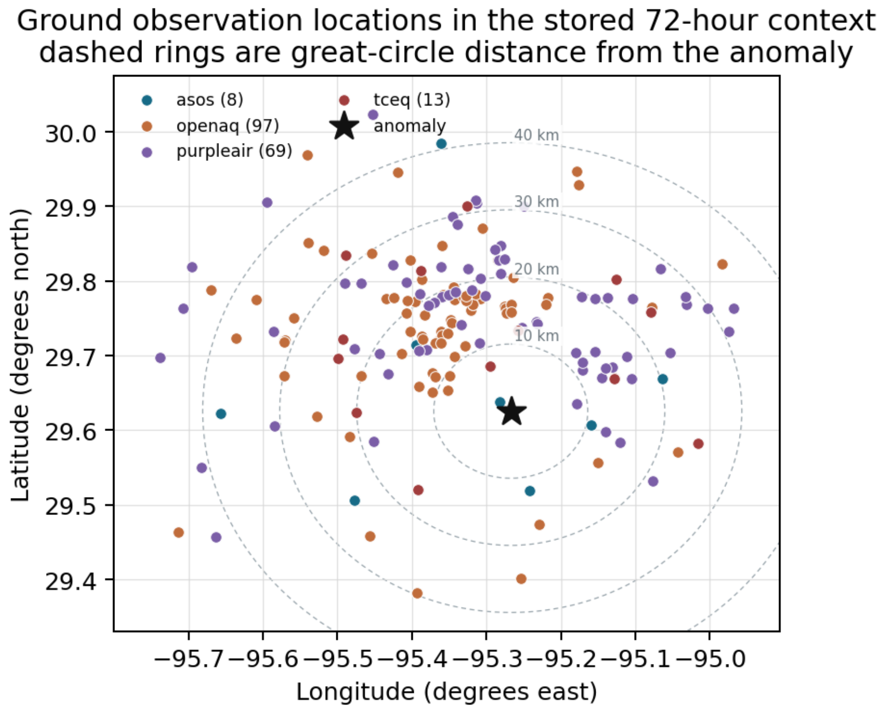
Source	Metric	Values
noaa_gfs	v_10m	07-21 12Z 0.3225, 18Z -0.6254, 07-22 00Z -0.0875, 06Z 1.269, 12Z -0.4623, 18Z -2.254, 07-23 00Z -0.004115, 06Z -2.975, 12Z -3.426, 18Z -5.142, 07-24 00Z 4.632, 06Z 3.39
openaq	ozone	07-21 06Z 0.006526, 12Z 0.01157, 18Z 0.0444, 07-22 00Z 0.02872, 06Z 0.006479, 12Z 0.01595, 18Z 0.06592, 07-23 00Z 0.03939, 06Z 0.02738, 12Z 0.02727, 18Z 0.02748, 07-24 00Z 0.02119, 06Z 0.01615
openaq	pm10	07-21 06Z 39.3, 12Z 41.33, 18Z 49.09, 07-22 00Z 46.83, 06Z 42.2, 12Z 45.33, 18Z 51.67, 07-23 00Z 55, 06Z 30.56, 12Z 25.22, 18Z 16.39, 07-24 00Z 19.61, 06Z 36.67
openaq	pm25	07-21 06Z 7.342, 12Z 6.97, 18Z 5.744, 07-22 00Z 5.904, 06Z 5.992, 12Z 6.323, 18Z 9.12, 07-23 00Z 10.54, 06Z 17.97, 12Z 17.84, 18Z 8.749, 07-24 00Z 7.979, 06Z 13.67
openweather	cloud_cover	07-21 06Z 0, 12Z 1.138, 18Z 1.806, 07-22 00Z 3.267, 06Z 21.53, 12Z 11.6, 18Z 56.93, 07-23 00Z 76.31, 06Z 66.06, 12Z 96.9, 18Z 96.73, 07-24 00Z 93.47, 06Z 70
openweather	humidity	07-21 06Z 90.54, 12Z 79.59, 18Z 48, 07-22 00Z 61.2, 06Z 79.5, 12Z 70.5, 18Z 49.03, 07-23 00Z 70.03, 06Z 76.65, 12Z 77.23, 18Z 77, 07-24 00Z 93.63, 06Z 93.5
openweather	precipitation	07-21 06Z 0, 12Z 0, 18Z 0, 07-22 00Z 0, 06Z 0, 12Z 0, 18Z 0.006, 07-23 00Z 1.453, 06Z 0, 12Z 0, 18Z 0.3917, 07-24 00Z 2.407, 06Z 0.3175
openweather	pressure	07-21 06Z 1012, 12Z 1013, 18Z 1011, 07-22 00Z 1010, 06Z 1010, 12Z 1011, 18Z 1008, 07-23 00Z 1009, 06Z 1010, 12Z 1010, 18Z 1009, 07-24 00Z 1012, 06Z 1014
openweather	temperature	07-21 06Z 25.62, 12Z 28.79, 18Z 36.46, 07-22 00Z 31.98, 06Z 27.45, 12Z 31.06, 18Z 38.97, 07-23 00Z 32.77, 06Z 29.84, 12Z 29.6, 18Z 30.21, 07-24 00Z 25.83, 06Z 25.38
openweather	wind_direction	07-21 06Z 206, 12Z 249.6, 18Z 259.8, 07-22 00Z 234.3, 06Z 189.3, 12Z 255.2, 18Z 179.6, 07-23 00Z 208.2, 06Z 107.8, 12Z 59.77, 18Z 106.7, 07-24 00Z 158.7, 06Z 152.2
openweather	wind_speed	07-21 06Z 2.252, 12Z 2.04, 18Z 1.646, 07-22 00Z 2.314, 06Z 1.842, 12Z 1.942, 18Z 2.202, 07-23 00Z 2.64, 06Z 2.198, 12Z 3.3, 18Z 4.267, 07-24 00Z 2.437, 06Z 2.183
purpleair	pm25	07-21 06Z 9.71, 12Z 8.546, 18Z 7.787, 07-22 00Z 9.643, 06Z 6.9, 12Z 8.044, 18Z 11.19, 07-23 00Z 14.58, 06Z 22.74, 12Z 17.97, 18Z 10.07, 07-24 00Z 10.46, 06Z 15.48
sentinel5p	s5p_aer_ai_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_ch4_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_co_column	07-21 18Z 0.02656, 07-22 18Z 0.03123, 07-23 18Z 0.02345
sentinel5p	s5p_co_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_hcho_column	07-21 18Z 0.0002243, 07-22 18Z 0.0003967, 07-23 18Z 0.0001668
sentinel5p	s5p_hcho_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_no2_column	07-21 18Z 5.873e-05, 07-22 18Z 5.036e-05
sentinel5p	s5p_no2_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_o3_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
sentinel5p	s5p_so2_column	07-21 18Z 3.63e-05, 07-22 18Z -2.412e-06, 07-23 18Z 5.971e-05
sentinel5p	s5p_so2_granule_available	07-21 18Z 1, 07-22 18Z 1, 07-23 18Z 1
tceq	co	07-21 06Z 0.1267, 12Z 0.1556, 18Z 0.1056, 07-22 00Z 0.21, 06Z 0.1056, 12Z 0.1556, 18Z 0.1222, 07-23 00Z 0.18, 06Z 0.1333, 12Z 0.1333, 18Z 0.1611, 07-24 00Z 0.22, 06Z 0.1167
tceq	no2	07-21 06Z 5.342, 12Z 5.402, 18Z 4.761, 07-22 00Z 10.57, 06Z 6.367, 12Z 7.011, 18Z 6.311, 07-23 00Z 7.105, 06Z 5.303, 12Z 6.175, 18Z 6.863, 07-24 00Z 7.503, 06Z 4.546
tceq	so2	07-21 06Z -0.28, 12Z -0.275, 18Z -0.1611, 07-22 00Z -0.1222, 06Z -0.2889, 12Z -0.07222, 18Z 0.07778, 07-23 00Z -0.1667, 06Z -0.3, 12Z -0.2333, 18Z -0.3278, 07-24 00Z -0.27, 06Z -0.35

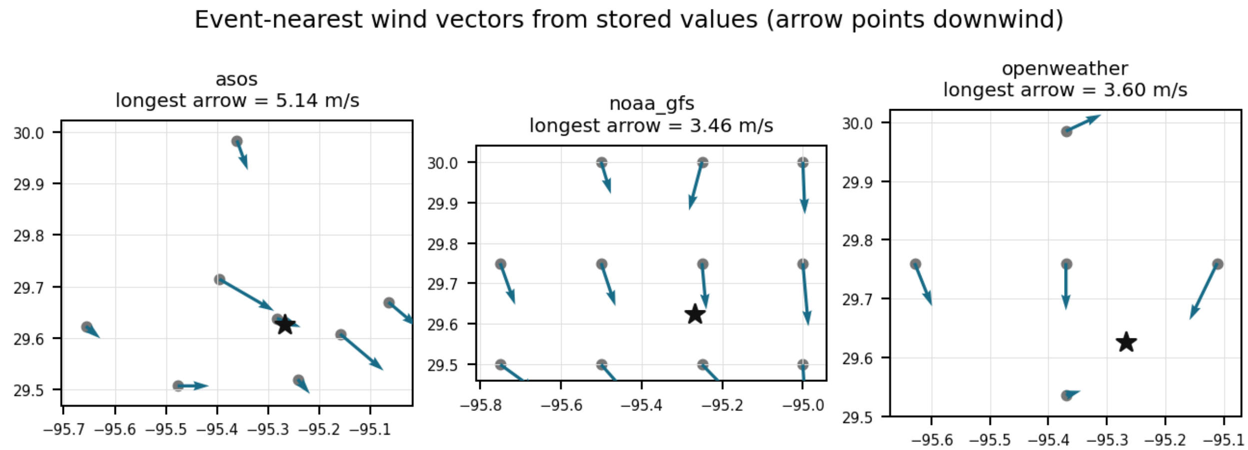
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	HOU (1.977 km) 71.27, EFD (10.684 km) 70.67, LVJ (12.118 km) 75.3, MCJ (15.782 km) 67.2, T41 (20.217 km) 72.29, AXH (24.247 km) 71.91, SGR (37.632 km) 74.22, IAH (40.904 km) 71.46
asos	temperature	HOU (1.977 km) 30.06, EFD (10.684 km) 30.31, LVJ (12.118 km) 29.3, MCJ (15.782 km) 30.59, T41 (20.217 km) 29.96, AXH (24.247 km) 29.46, SGR (37.632 km) 29.73, IAH (40.904 km) 30.11
asos	wind_direction	HOU (1.977 km) 183.1, EFD (10.684 km) 180.9, LVJ (12.118 km) 165.3, MCJ (15.782 km) 197.6, T41 (20.217 km) 190.3, AXH (24.247 km) 144.1, SGR (37.632 km) 196.7, IAH (40.904 km) 199.8
asos	wind_speed	HOU (1.977 km) 3.58, EFD (10.684 km) 3.451, LVJ (12.118 km) 1.993, MCJ (15.782 km) 4.13, T41 (20.217 km) 3.183, AXH (24.247 km) 1.612, SGR (37.632 km) 2.929, IAH (40.904 km) 3.294
noaa_gfs	gh_500	gfs:29.75,-95.25 (13.932 km) 5934, gfs:29.50,-95.25 (14.065 km) 5931, gfs:29.75,-95.50 (26.402 km) 5935, gfs:29.50,-95.50 (26.496 km) 5932, gfs:29.75,-95.00 (29.284 km) 5934, gfs:29.50,-95.00 (29.376 km) 5931, gfs:30.00,-95.25 (41.664 km) 5937, gfs:30.00,-95.50 (47.304 km) 5938, +3 more
noaa_gfs	pbl_height	gfs:29.75,-95.25 (13.932 km) 863.1, gfs:29.50,-95.25 (14.065 km) 556.4, gfs:29.75,-95.50 (26.402 km) 937.4, gfs:29.50,-95.50 (26.496 km) 542.5, gfs:29.75,-95.00 (29.284 km) 591.1, gfs:29.50,-95.00 (29.376 km) 538.8, gfs:30.00,-95.25 (41.664 km) 797.6, gfs:30.00,-95.50 (47.304 km) 852.3, +3 more
noaa_gfs	precipitable_water	gfs:29.75,-95.25 (13.932 km) 49.3, gfs:29.50,-95.25 (14.065 km) 50.31, gfs:29.75,-95.50 (26.402 km) 48.36, gfs:29.50,-95.50 (26.496 km) 48.96, gfs:29.75,-95.00 (29.284 km) 50.18, gfs:29.50,-95.00 (29.376 km) 50.8, gfs:30.00,-95.25 (41.664 km) 49.34, gfs:30.00,-95.50 (47.304 km) 48.78, +3 more
noaa_gfs	surface_pressure	gfs:29.75,-95.25 (13.932 km) 1009, gfs:29.50,-95.25 (14.065 km) 1009, gfs:29.75,-95.50 (26.402 km) 1008, gfs:29.50,-95.50 (26.496 km) 1008, gfs:29.75,-95.00 (29.284 km) 1010, gfs:29.50,-95.00 (29.376 km) 1010, gfs:30.00,-95.25 (41.664 km) 1007, gfs:30.00,-95.50 (47.304 km) 1006, +3 more
noaa_gfs	t_850	gfs:29.75,-95.25 (13.932 km) 21.33, gfs:29.50,-95.25 (14.065 km) 21.15, gfs:29.75,-95.50 (26.402 km) 21.34, gfs:29.50,-95.50 (26.496 km) 21.25, gfs:29.75,-95.00 (29.284 km) 21.3, gfs:29.50,-95.00 (29.376 km) 21.12, gfs:30.00,-95.25 (41.664 km) 21.36, gfs:30.00,-95.50 (47.304 km) 21.38, +3 more
noaa_gfs	u_10m	gfs:29.75,-95.25 (13.932 km) 0.4448, gfs:29.50,-95.25 (14.065 km) 0.7273, gfs:29.75,-95.50 (26.402 km) 0.5581, gfs:29.50,-95.50 (26.496 km) 1.54, gfs:29.75,-95.00 (29.284 km) 0.3923, gfs:29.50,-95.00 (29.376 km) 1.151, gfs:30.00,-95.25 (41.664 km) 0.1931, gfs:30.00,-95.50 (47.304 km) 0.5606, +3 more
noaa_gfs	v_10m	gfs:29.75,-95.25 (13.932 km) -0.5792, gfs:29.50,-95.25 (14.065 km) -0.1709, gfs:29.75,-95.50 (26.402 km) -0.6542, gfs:29.50,-95.50 (26.496 km) -0.3001, gfs:29.75,-95.00 (29.284 km) -0.1517, gfs:29.50,-95.00 (29.376 km) 0.4249, gfs:30.00,-95.25 (41.664 km) -0.9667, gfs:30.00,-95.50 (47.304 km) -0.7442, +3 more
openaq	ozone	320 (0.0 km) 0.02815, 325 (7.264 km) 0.009182, 3378 (12.079 km) 0.02825, 332 (14.284 km) 0.03371, 2039 (16.472 km) 0.02512, 3055 (16.85 km) 0.02897, 312 (20.009 km) 0.03097, 1319333 (20.557 km) 0.03228, +9 more
openaq	pm10	13380082 (12.079 km) 34.43, 271 (14.284 km) 44.44, 15852419 (16.681 km) 30.83
openaq	pm25	14157975 (0.003 km) 9.895, 14140751 (7.245 km) 9.045, 15472101 (8.787 km) 6.081, 13418566 (9.543 km) 8.084, 8967021 (10.575 km) 9.264, 16564662 (10.96 km) 9.854, 9416627 (11.119 km) 5.973, 15990898 (11.396 km) 9.823, +69 more
openweather	cloud_cover	grid:south (14.076 km) 50.92, grid:center (17.969 km) 46.4, grid:east (21.261 km) 44.26, grid:west (37.982 km) 43.51, grid:north (41.171 km) 37.72
openweather	humidity	grid:south (14.076 km) 73.29, grid:center (17.969 km) 70.71, grid:east (21.261 km) 77.22, grid:west (37.982 km) 71.47, grid:north (41.171 km) 70.83
openweather	precipitation	grid:south (14.076 km) 0.404, grid:center (17.969 km) 0.2092, grid:east (21.261 km) 0.2751, grid:west (37.982 km) 0.6304, grid:north (41.171 km) 0.2531
openweather	pressure	grid:south (14.076 km) 1010, grid:center (17.969 km) 1010, grid:east (21.261 km) 1010, grid:west (37.982 km) 1011, grid:north (41.171 km) 1011
openweather	temperature	grid:south (14.076 km) 30.36, grid:center (17.969 km) 31.35, grid:east (21.261 km) 30.13, grid:west (37.982 km) 30.92, grid:north (41.171 km) 30.88
openweather	wind_direction	grid:south (14.076 km) 188.1, grid:center (17.969 km) 196.7, grid:east (21.261 km) 183.1, grid:west (37.982 km) 188.8, grid:north (41.171 km) 162
openweather	wind_speed	grid:south (14.076 km) 2.457, grid:center (17.969 km) 2.373, grid:east (21.261 km) 3.235, grid:west (37.982 km) 1.678, grid:north (41.171 km) 2.357

Source	Metric	Values
purpleair	pm25	182191 (8.594 km) 12.79, 6752 (10.934 km) 8.051, 186263 (11.178 km) 13.45, 244939 (11.797 km) 10.45, 235425 (12.183 km) 11.68, 99797 (12.499 km) 10.68, 247283 (12.722 km) 12.7, 128007 (12.806 km) 6.247, +61 more
tceq	co	482011035 (12.059 km) 0.1672, 482011039 (14.272 km) 0.09538, 482011052 (24.007 km) 0.1682
tceq	no2	482010416 (7.255 km) 6.567, 482011035 (12.059 km) 7.012, 482011039 (14.272 km) 7.495, 480391004 (16.839 km) 4.888, 482011015 (23.434 km) 8.597, 482010055 (23.737 km) 3.812, 482010026 (23.979 km) 7.075, 482011052 (24.007 km) 3.948, +4 more
tceq	so2	482011035 (12.059 km) -0.4794, 482011039 (14.272 km) 0.3185, 482010051 (20.024 km) -0.4762





Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO_2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

High PM2.5 levels were detected in the area, with values exceeding 12 ug/m3 at multiple sensors.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

Hot, relatively dry afternoon conditions favored rapid photochemical ozone production at the anomaly time because near-surface temperature was about 38 to 39 degrees Celsius and humidity was about 38 to 50 percent near 19Z on 2026-07-22, coincident with the severe ozone peak.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

The observed ozone concentration of 0.081 ppm at 19:00:00+00:00 on July 22, 2026, represents a statistically significant deviation with a z-score of 5.57 above expected levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

Ground monitoring sensors recorded a severe surface ozone concentration spike of 0.081 ppm at 19:00 UTC on July 22, 2026, relative to an expected baseline of approximately 0.015 ppm.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

The openaq ozone anomaly at 29.626, -95.267 had z-score 5.567538938752313 and severity severe, and the surrounding 6-hour mean ozone for 2026-07-22 18Z was 0.06592 ppm across 17 openaq sensors within 50 km.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

The openaq ozone anomaly at 29.626, -95.267 in Houston occurred at 2026-07-22T19:00:00+00:00 with a measured ozone value of 0.081 ppm.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

A deep afternoon boundary layer supported vertical mixing and surface expression of ozone because the modeled planetary boundary layer height near the anomaly reached about 2017 meters at 18Z on 2026-07-22, much higher than the nighttime and morning values in the same window.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

High ambient surface temperatures reaching 38 to 39 degrees Celsius combined with low relative humidity around 38 to 50 percent promoted intense photochemical ozone formation in the Houston area.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

A fresh primary combustion or sulfur plume was not the dominant cause of the ozone peak because concurrent NO2, SO2, CO, and PM2.5 measurements did not show a comparably sharp co-located spike at the anomaly time.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.89 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.5349815129739817 m/s; n=360 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

The air quality anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter and volatile organic compounds.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

Surface ambient temperatures reached approximately 38 degrees Celsius alongside low relative humidity near 38 percent around 19:00 UTC on July 22, 2026, creating optimal meteorological conditions for photochemical ozone formation.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

Mesoscale transport and partial stagnation likely helped concentrate or advect an ozone-rich air mass over east-central Houston because winds were light to modest and variable across local observations while satellite CO was elevated without a corresponding SO2 enhancement.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.89 m/s, below its cutoff of 1.5 m/s (mean - 2*SD gave -0.5349815129739817 m/s; n=360 wind readings). Cutoff floor: confirmed 2026-07-24.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

A severe ozone anomaly was recorded in Houston at coordinates (29.626, -95.267) on July 22, 2026, at 19:00:00+00:00, reaching a concentration of 0.081 ppm compared to an expected baseline of approximately 0.0155 ppm.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

The anomaly occurred on July 22nd, with the highest PM2.5 levels recorded at 19:00:00+00:00.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

Strong local photochemical ozone production was likely promoted by extreme afternoon heat, low relative humidity, and a deep daytime boundary layer near the anomaly time in Houston.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

The PM2.5 levels are elevated, with a mean of 11.5612 ug/m3 and a maximum value of 149.3 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

Unusual wind patterns were observed, including a significant increase in wind speed and direction changes, which may have contributed to the air quality anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

High surface temperatures reaching 38.0 deg C recorded by ASOS and low relative humidity recorded by OpenWeather created meteorologically favorable conditions for intense daytime photochemical ozone formation on July 22, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

Elevated VOC-driven chemical reactivity was likely a major contributor because formaldehyde column values were unusually high near the anomaly time while surface NO2 remained only moderate, a pattern consistent with efficient ozone formation in a VOC-rich urban-industrial environment.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

Satellite observations detected a pronounced peak in vertical column formaldehyde density over Houston on July 22, 2026, indicating elevated volatile organic compound precursor levels during the ozone event.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

The Houston ozone anomaly on 2026-07-22 19:00 UTC was most likely driven by unusually strong afternoon photochemical ozone production under very hot and relatively dry conditions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

OpenAQ surface monitors recorded a peak ozone concentration of 0.081 ppm at 19:00 UTC on July 22, 2026, coinciding with daytime boundary layer growth modeled by GFS.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

Light surface wind speeds averaging under 3 m/s facilitated local accumulation of precursor emissions and limited advective ventilation across the urban domain.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

The expected openaq ozone value for 2026-07-22T19:00:00+00:00 at 29.626, -95.267 was 0.015450980392156862 ppm, so the observed ozone value of 0.081 ppm was much higher than expected.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

The air quality anomaly in Houston, Texas on July 22nd was caused by a combination of factors including high PM2.5 levels and unusual wind patterns.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

Coinciding with the peak ozone observation on July 22, 2026, surface weather observations near the anomaly recorded high ambient temperatures up to 38.0 degC and decreased relative humidity down to 37.58 percent at 18:55 UTC.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

Reactive precursor availability is supported more strongly than a fresh combustion plume because satellite HCHO and CO were elevated near 18-19Z on 2026-07-22 while local NO2, SO2, PM2.5, and CO remained modest at the surface near the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

Elevated volatile organic compound levels, indicated by increased atmospheric formaldehyde column densities measured by satellite, provided reactive precursors for rapid midday ozone synthesis.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

Sentinel-5P satellite measurements showed an elevated formaldehyde column density of 0.000397 mol/m^2 on July 22, indicating enhanced volatile organic compound precursor levels during the peak ozone event.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

The anomaly is due to a combination of factors including high temperatures, humidity, and stagnant atmospheric conditions, which have led to the formation of ground-level ozone and other pollutants.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

A severe thunderstorm or tornado event in the Houston area has stirred up dust and debris, contributing to the air quality anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

Reactive precursor availability was consistent with VOC-influenced ozone formation because sentinel5p HCHO and CO were elevated near the anomaly time while surface ozone simultaneously rose well above its local baseline.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.